Low cost.	More costlier instrument.
The output is ambiguous in many times.	Unambiguous reading due to digital indication.

10.5 DIGITAL FREQUENCY METER

A frequency counter is a digital instrument that can measure and display the frequency of any periodic waveform. It operates on the principle of gating the unknown input signal into the counter for a predetermined time. For example, if the unknown input signal were gated into the counter for exactly 1 second, the number of counts allowed into the counter would be precisely the frequency of the input signal. The term *gated* comes from the fact that an AND or an OR gate is employed for allowing the unknown input signal into the counter to be accumulated.

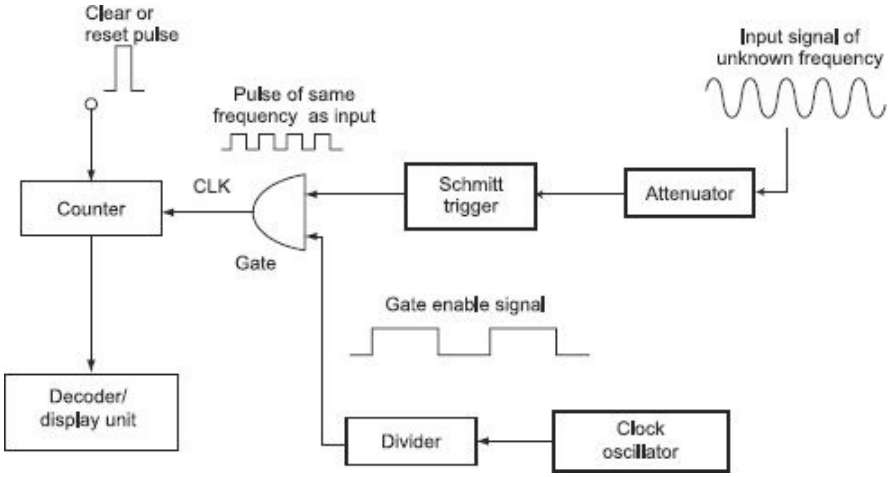


Figure 10.4 *Block diagram of frequency counter*

One of the most straightforward methods of constructing a frequency counter is shown in Figure 10.4 in simplified form. It consist of a counter with its associated display/decoder circuitry, clock oscillator, a divider and an AND gate. The counter is usually made up of cascaded Binary Coded Decimal (BCD) counters and the display/decoder unit converts the BCD outputs into a decimal display for easy monitoring. A GATE ENABLE signal of known time period is generated with a clock oscillator and a divider circuit and is applied to one leg of an AND gate. The unknown signal is applied to the other leg of the AND gate and acts as the clock for the counter. The counter advances one count for each transition of the unknown signal, and at the end of the known time interval, the contents of the counter will be equal to the number of periods of the unknown input signal that have occurred during time interval, *t*. In other words, the counter contents will be proportional to the frequency of the unknown input signal. For instance if the gate signal is of a time of exactly 1 second and the unknown input signal is a 600-Hz square wave, at the end of 1 second the counter will counts up to 600, which is exactly the frequency of the unknown input signal.

The waveform in Figure 10.5 shows that a clear pulse is applied to the counter at t_0 to set the counter at zero. Prior to t_1, the GATE ENABLE signal is LOW, and so the output of the AND gate will be LOW and the counter will not be counting. The GATE ENABLE

goes HIGH from t_1 tot_2 and during this time interval t (= $t_2 - t_1$), the unknown input signal pulses will pass through the AND gate and will be counted by the counter. After t_2, the AND gate output will be again LOW and the counter will stop counting. Thus, the counter will have counted the number of pulses that occurred during the time interval, t of the GATE ENABLE SIGNAL, and the resulting contents of the counter are a direct measure of the frequency of the input signal.

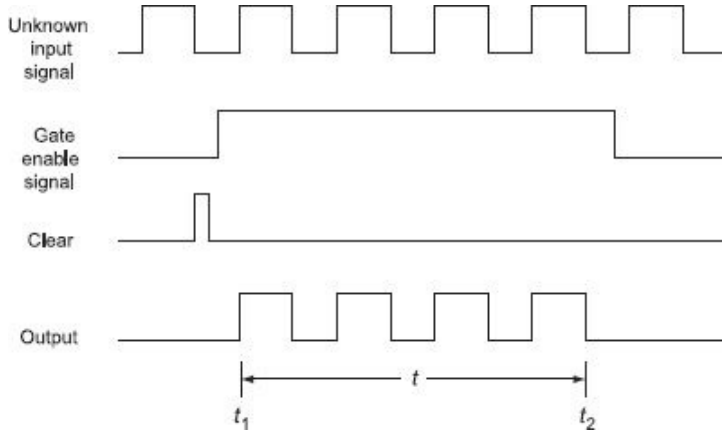


Figure 10.5 *Different waveforms in a frequency counter*

The accuracy of the measurement depends almost entirely on the time interval of the GATE ENABLE signal, which needs to be controlled very accurately. A commonly used method for obtaining very accurate GATE ENABLE signal is shown in Figure 10.6. A crystal controlled oscillator is employed for generating a very accurate 100 kHz waveform, which is shaped into the square pulses and fed to a series of decade counters that are being used to successively divide this 100 kHz frequency by 10. The frequencies at the outputs of each decade counter are as accurate as the crystal frequency.

The switch is used to select one of the decade counter output frequencies to be supplied to a single Flip-flop to be divided by 2. For instance in switch position 1, the 1 Hz pulses are supplied to flip-flop Q, which acts as a toggle flip-flop so that its output will be a square wave with a period if T = 2 s and a Tpulse duration,$t = \frac{T}{2} = 1\,\text{s}$ 1s . In position 2, the pulse duration would be 0.1s, and so on in other positions of the select switch.

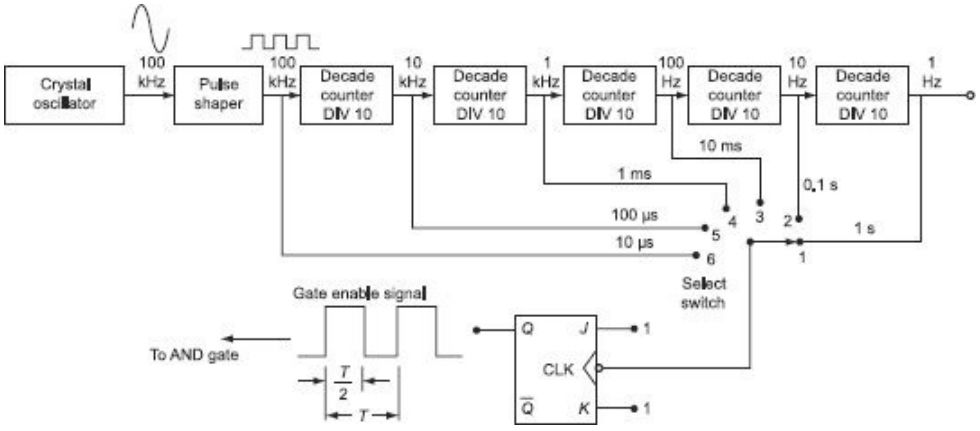


Figure 10.6 *Method of obtaining very accurate GATE ENABLE signal*

DIGITAL VOLTMETERS (DVMS)

10.6

The Digital Voltmeter (DVM) displays measurement of ac or dc voltages as discrete numbers instead of a pointer deflection on a continuous scale as in analog instruments. It is a versatile and accurate instrument that is employed in many laboratory measurement applications. Because of development and perfection of IC modules, their size, power consumptions and cost of the digital voltmeters has been drastically reduced and, therefore, DVMs are widely used for all measurement purposes.

The block diagram of a simple digital voltmeter is shown in Figure 10.7. The unknown signal is fed to the pulse generator which generates a pulse whose width is directly proportional to the input unknown voltage.

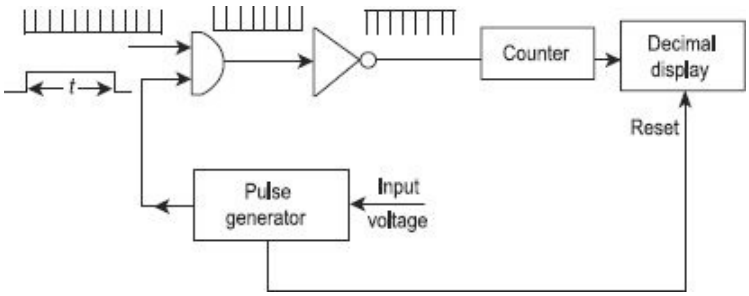


Figure 10.7 *Block diagram of DVM*

The output of the pulse generator is applied to one leg of an AND gate. The input signal to the other leg of the AND gate is a train of pulses. The output of the AND gate is, thus, a positive trigger train of duration *t* second and the inverter converts it into a negative trigger train. The counter counts the number of triggers in *t* seconds which is proportional to the voltage under measurement. Thus, the counter can be calibrated to indicate voltage in volts directly.

Thus, the DVMs described above is an Analog to Digital Converter (ADC) which converts an analog signal into a train of pulses, the number of which is proportional to the input voltage. So a digital voltmeter can be made by using any one of the analog to digital conversion methods and can be represented by a block diagram shown in Figure 10.8. So the DVMs can be classified on the basis of ADCs used.

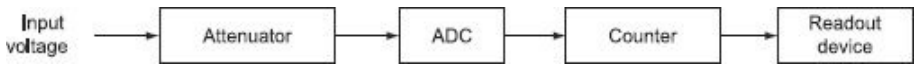


Figure 10.8 *Representation of DVM using blocks*

The input range of the DVM may vary from ±1.00000 V to ±1000.00 V and its limiting accuracy is as high as ±0.005 percent of the reading. Its resolution may be 1 part in 10^6, giving 1 μV reading of the 1 V input range. It has high input resistance of the order of 10 MΩ and input capacitance of the order of 40 pF.

Digital voltmeters employing different analog to digital conversion methods are described below:

10.6.1 Ramp-Type DVM

The operation of a ramp-type DVM is based on the measurement of the time that a linear

Electrical Drives and Instrumentation

Display Device

Introduction

- Display devices are the output devices for presentation of information in text or image form
- For displaying the information in an appropriate manner these devices must be controlled by some other external devices
- Some displays can show digits and alphanumeric characters only. Some displays can show images and all type of characters
- Most commonly used displays along with microcontrollers are LEDs, LCD, GLCD, and 7-segment displays

Display Using LED

- Light emitting diode (LED) is the most commonly used device for displaying the status of microcontroller pins
- These display devices are commonly used for the indication of alarms, inputs and timers
- There are two ways by we can connect LEDs to microcontroller unit
- Those two ways are active high logic and active low logic
- Active high logic means LED will be ON when port pin is 1 and LED will be OFF when pin is 0

7-Segment LED Display

- 7-Segment LED display can be used for displaying digits and few characters
- A seven segment display consists of 7 LEDs arranged in the form of Square ‘8’ and a single LED as dot character
- A 7 seven segment display is an electronic display, which displays 0-9 digital information
- They are available in common cathode mode and common anode mode
- There are state lines in LED, anode is given to positive terminal and cathode is given to negative terminal then LED will glow

7-Segment LED Display

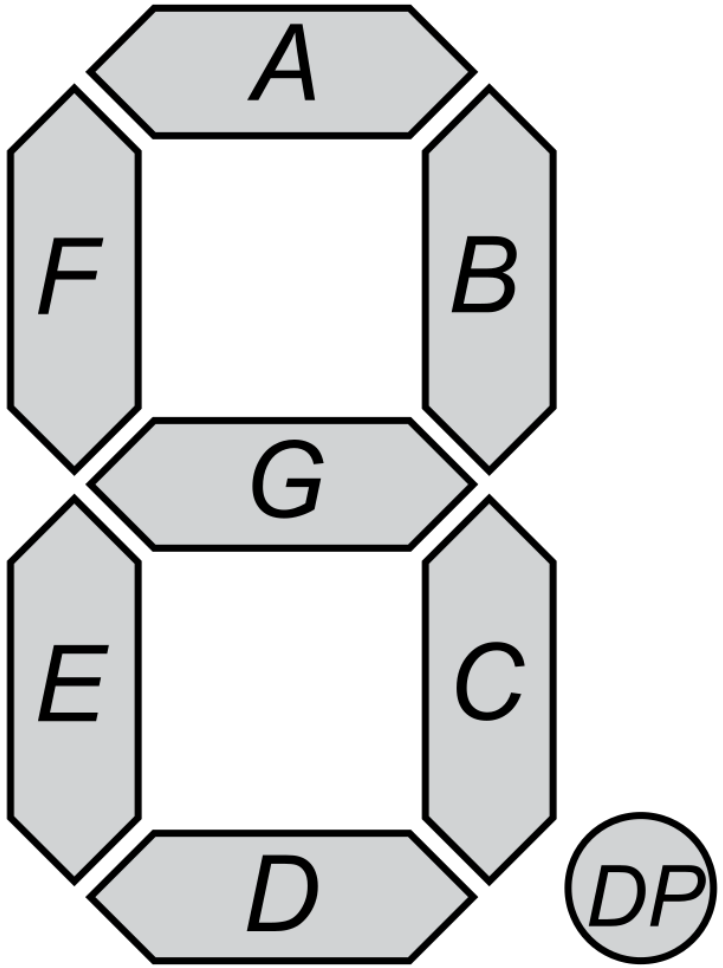
- In common cathode, the negative terminals of all LEDs are connected to the common pins to ground and a particular LED glows when its corresponding pin is given high
- The cathodes of all LEDs are connected together to a single terminal and the anodes of all LEDs are left alone
- In common anode arrangement, the common pin is given a high logic and the LED pins are given low to display a number
- In common anode, all the anodes are connected together and all the cathodes are left alone

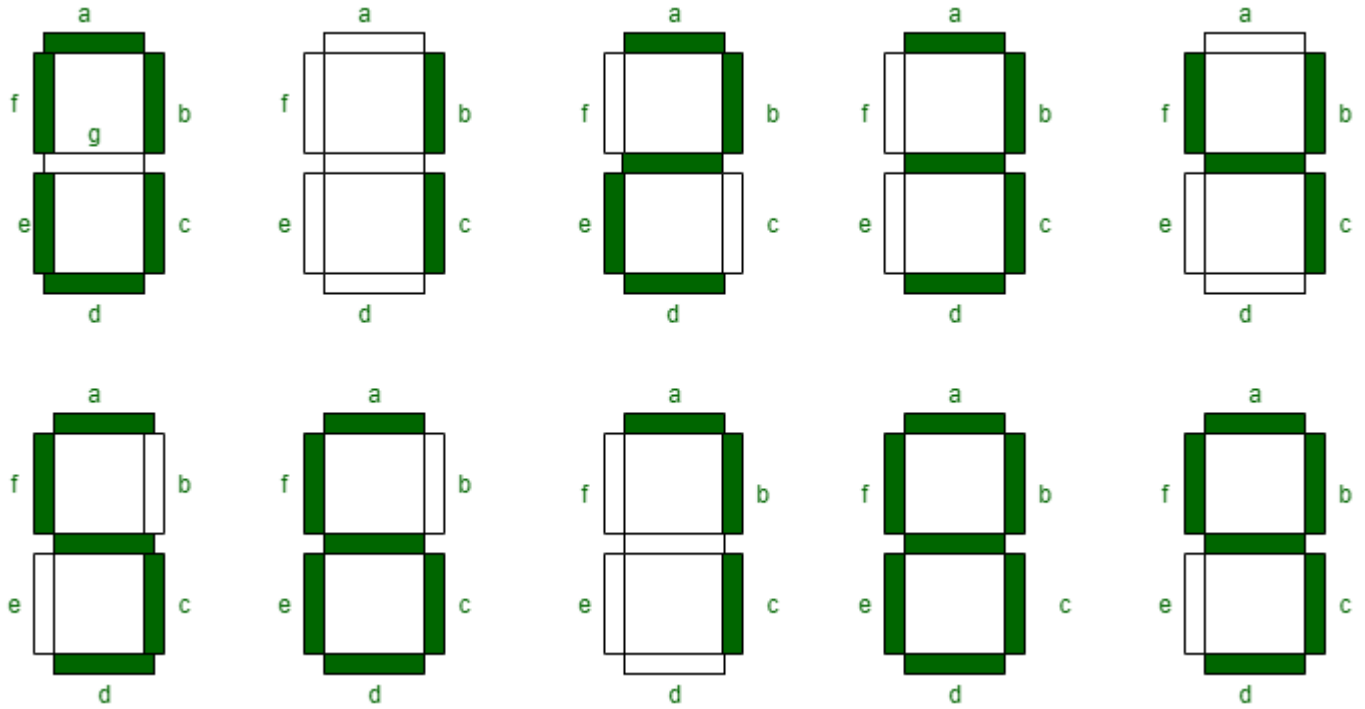
7-Segment LED Display

- Thus when we gives first signal is high or 1 then only there is a lean in display if not there is no lean in display

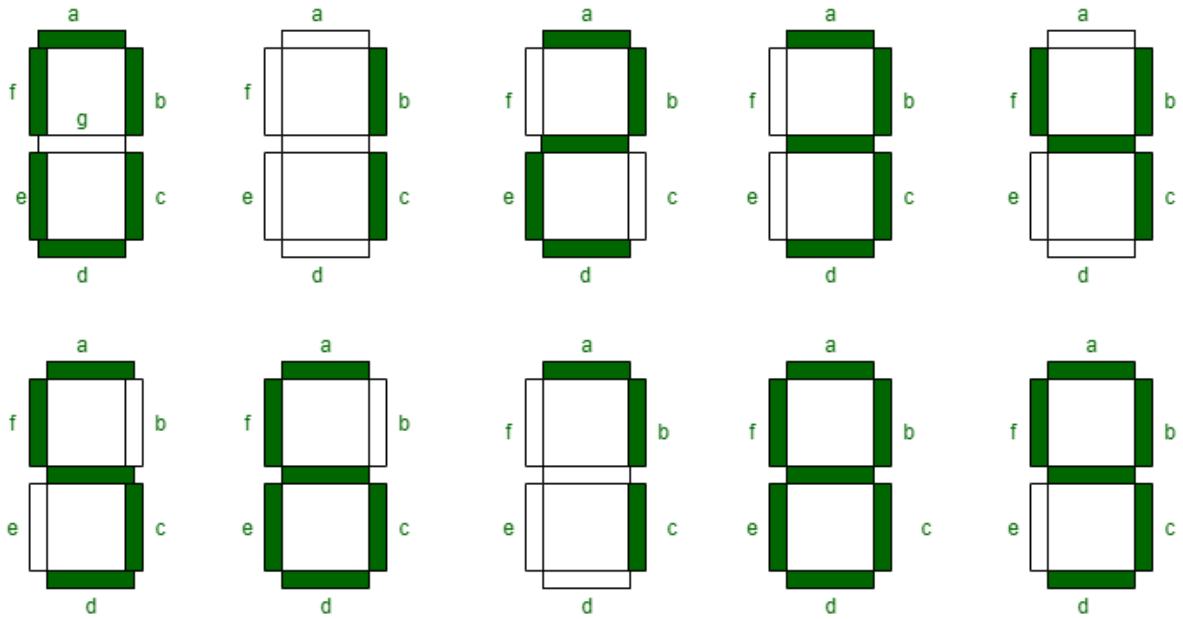
Binary Inputs				Decoder Outputs							7-Segment Display Outputs
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0	1	2
0	0	1	1	1	1	1	1	0	0	1	3
0	1	0	0	0	1	1	0	0	1	1	4
0	1	0	1	1	0	1	1	0	1	1	5
0	1	1	0	1	0	1	1	1	1	1	6
0	1	1	1	1	1	1	0	0	0	0	7
1	0	0	0	1	1	1	1	1	1	1	8
1	0	0	1	1	1	1	1	0	1	1	9

7-Segment LED Display





7-Segment LED Display

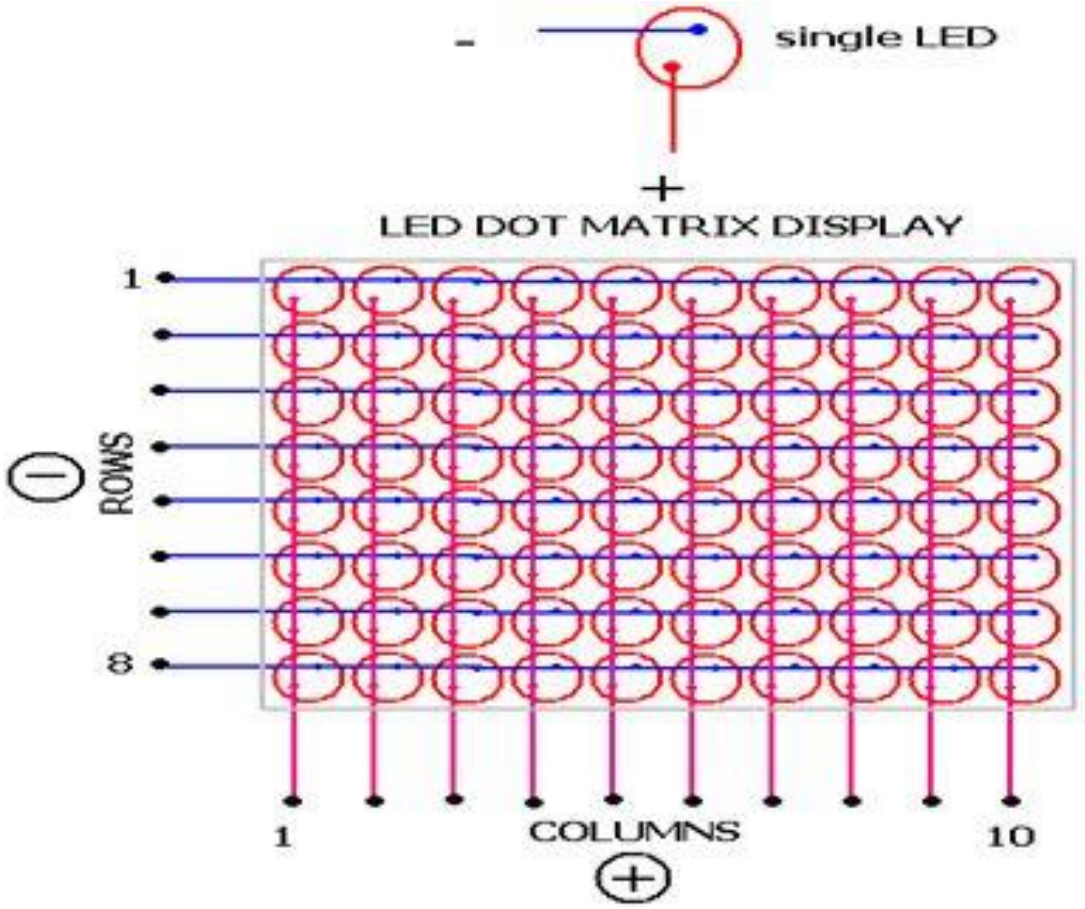


Binary Inputs				Decoder Outputs							7-Segment Display Outputs
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	1	1	1	1	1	1	0	0
0	0	0	1	0	1	1	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0	1	2
0	0	1	1	1	1	1	1	0	0	1	3
0	1	0	0	0	1	1	0	0	1	1	4
0	1	0	1	1	0	1	1	0	1	1	5
0	1	1	0	1	0	1	1	1	1	1	6
0	1	1	1	1	1	1	0	0	0	0	7
1	0	0	0	1	1	1	1	1	1	1	8
1	0	0	1	1	1	1	1	0	1	1	9

Dot Matrix LED Display

- Dot matrix LED display contains the group of LEDs as a two dimensional array
- They can display different types of characters or a group of characters
- Dot matrix display is manufactured in various dimensions
- Row anode-column cathode or Row cathode-column anode
- By using this dot matrix display we can reduce the number of pins required for controlling all the LEDs
- Each LED will be addressed by its row and column number

Dot Matrix LED Display



Controlling the LED Matrix

- Since all the LEDs in a matrix share their positive and negative terminals in each row and column, it is not possible controlling of each LED at the same time
- The matrix controlled through each row very quickly by triggering the correct column pins to light the desired LED's for that particular row
- If the switching done with a fixed rate, humans can't see the displaying message, because human eye can't detect the images with in the milliseconds of time
- Thus the displaying of a message on LED matrix must be controlled, with the rows being scanned sequentially at a rate greater than 40 MHz while sending out the column data at the exact same rate

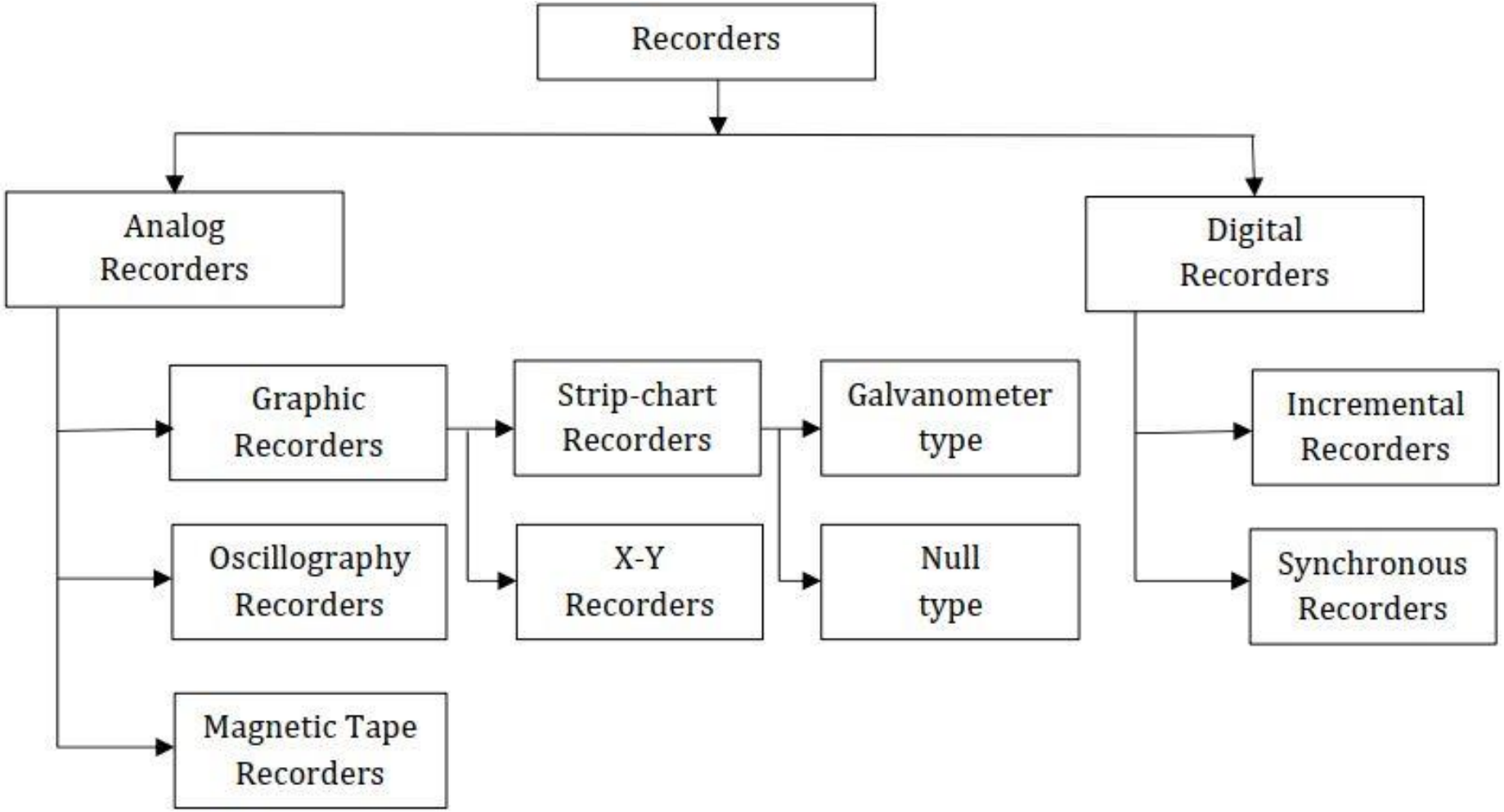
Electrical Drives and Instrumentation

Recorder

Introduction

- A recorder records electrical and non – electrical quantities as a function of time
- Currents and voltages can be recorded directly while the non – electrical quantities are recorded indirectly by first converting them to equivalent currents or voltages with the help of sensors or transducers

Classification of Recorder



Classification of Recorder

Analog recording:

- An analog recording is one where a property or characteristic of a physical recording medium is made to vary in a manner analogous to the variations in air pressure of the original sound.
- Generally, the air pressure variations are first converted (by a transducer such as a microphone) into an electrical analog signal in which either the instantaneous voltage or current is analogous to the instantaneous air pressure.
- The variations of the electrical signal in turn are converted to variations in the recording medium by a recording machine such as a tape recorder or cutting lathe.
- Example – PMMC instrument (**P**ermanent **M**agnet **M**oving **C**oil)

Classification of Recorder

Graphic recording:

- Graphic recorders generally are devices which display and store a pen and ink record of the history of some physical event.
- Basic elements of a recorder include a chart for displaying and storing the recorded information, a stylus moving in a proper relationship to the paper and suitable means of interconnection to couple the stylus to the source of information.

- The graphic recorders can be classified in two categories:

  (i) *Strip chart recorders:* A strip chart recorder records one or more variables with respect to time. It is an X-t recorder.

  (ii) *X-Y recorders:* An X-Y recorder records one or more dependent variables with respect to an independent variable.


Thank
you